

SPARSE CSP ALGORITHM VIA JOINT SPATIO-TEMPORAL FILTERING

Aimin Jiang*, Jing Shang*, Weigao Cheng*, Xiaofeng Liu*, Hon Keung Kwan†, and Yanping Zhu‡

*College of Internet of Things Engineering, Hohai University, Changzhou, Jiangsu, China

†Department of Electrical and Computer Engineering, University of Windsor, Windsor, Ontario, Canada

‡School of Information Science and Engineering, Changzhou University, Changzhou, Jiangsu, China

ABSTRACT

Common spatial pattern (CSP) is widely used in motor imagery classification tasks. Classical CSP depends only on spatial filters. To improve its performance, a novel and efficient spatio-temporal filtering strategy is proposed in this paper to extract discriminative features. Common temporal filters are shared among all the spatial channels, so as to reduce the overfitting risk in the case of a small sample size. An efficient alternating optimization algorithm is also developed to optimize coefficients of spatial and temporal filters. To alleviate adverse effects of noise and artifacts and improve implementation efficiency, an ℓ_1 -norm-based sparsity regularization term is further introduced. The resulting problem is tackled by the reweighting technique. The effectiveness of the proposed algorithm is validated by the experiments using open datasets of BCI Competition.

Index Terms— Alternating optimization, common spatial pattern (CSP), electroencephalograph (EEG), reweighting technique, sparsity, spatio-temporal filter.

1. INTRODUCTION

A brain-computer interface (BCI) system provides users with severe motor disabilities the possibility of a nonmuscular communication with a device by recognizing patterns of brain activities [1]-[4]. In a variety of BCI applications, electroencephalography (EEG) is the most effective and efficient way to measure brain activities [5]. Nevertheless, due to its low signal-to-noise ratio (SNR) and inherent nonstationarity of EEG signals, feature extraction is important to the success of EEG-based BCIs.

In motor imagery (MI) tasks, different brain states have to be accurately classified [6]-[7]. So far, common spatial pattern (CSP) [8]-[10] is considered as an effective way to

encode discriminative features in binary classification. Its objective is to design subject-dependent spatial filters, such that variances of spatially filtered EEG signals in different classes are simultaneously maximized and minimized. The CSP problem can be efficiently tackled by the generalized eigenvalue decomposition (EVD) in practice. Recent advances show that the performance of traditional CSP algorithm can be further enhanced by various techniques [11]-[15]. For example, besides spatial filters, temporal filters are also employed to process nonstationary EEG signals [11]-[12]. In [11], an augmented data matrix constructed by the original EEG signals and their time-delayed copies is used in the CSP algorithm. Temporal filters and the classifier are simultaneously optimized according to statistical learning theory to achieve good performance in [12]. The major drawback of joint spatio-temporal filtering is that the overfitting risk could be increased since more filter coefficients need to be optimized. This requires special techniques when dealing with a small amount of data samples.

Another strategy to enhance the performance of traditional CSP algorithm is by the regularization techniques. For example, the sparsity working along with CSP technique is able to identify and remove noisy EEG channels [13]-[15]. Sparse spatial CSP (ssCSP) based on ℓ_1/ℓ_2 norm is proposed in [15] to promote sparsity, while ℓ_1 norm is employed by the regularized CSP (rCSP) in [14] for channel selection. Also, filter sparsity can also be directly enhanced subject to an appropriate performance constraint [16]-[17]. In the above sparsity-inspired CSP models, the introduction of ℓ_1 norm or its variants generally lead to more complicated optimization problems. Accordingly, they require more computational costs than those of traditional CSP technique.

Generally, CSP algorithms rely on sample covariance matrices. Some approaches directly manipulate these symmetric and positive-definite (PSD) matrices via Riemannian geometry [18]-[19]. Based on Riemannian distance and mean, a multiclass classification method called minimum distance to riemannian mean (MDRM) proposed in [18] classify a unknown covariance matrix by measuring its distance to each intraclass covariance matrix. If covariance matrices are projected to the Riemannian tangent space, classical techniques (e.g., LDA [21] and SVM [22]) can be applied on vector-

This work was supported in part by the National Key Research and Development Program under grant 2018AAA0100800, the National Nature Science Foundation of China under grants 61772090, 61701471, and 61801055, the Fundamental Research Funds for the Central Universities of China under grants 2018B23014 and 2018B47114, and the Key Development Program of Jiangsu Province of China under grants BE2017071, BE2017647, and BE2018004-04.

ized features. Compared to CSP algorithms, manifold-based methods need more training data in practice.

In this paper, we still focus on the CSP technique using joint spatio-temporal filtering. But the same temporal filter is shared among all the EEG channels, such that the number of filter coefficients can be significantly reduced. Although the joint filtering problem is nonconvex, spatial and temporal filters can be alternatively updated, such that their respective design problems can still be formulated in the CSP form and thus efficiently solved by the generalized EVD. Furthermore, an ℓ_1 -norm regularization term is introduced into spatial and temporal filter design problems to encourage sparsity. To enhance computational efficiency, the reweighting technique is deployed, and the resulting problem can still be tackled by the classical CSP technique.

The paper is organized as follows. In Section II, we first present our CSP model based on joint spatio-temporal filtering. Then, an alternating optimization algorithm is further developed. The sparse filter design problems is finally considered in Section II. Experimental results are presented in Section III. Section IV concludes the paper.

2. PROPOSED CSP ALGORITHM BY SPATIO-TEMPORAL FILTERING

Let $\mathbf{X}_i = [\mathbf{x}_{i,1} \ \mathbf{x}_{i,2} \ \dots \ \mathbf{x}_{i,K}] \in \mathbb{R}^{C \times K}$ denote a data matrix of the i th trial, where C and K are the numbers of spatial channels and sampling points, respectively. The traditional CSP problem is formulated as

$$\max_{\mathbf{w} \in \mathbb{R}^C} \frac{\mathbf{w}^T \mathbf{P}_1 \mathbf{w}}{\mathbf{w}^T \mathbf{P}_2 \mathbf{w}} \quad (1)$$

where $\mathbf{w}$ is the coefficient vector of a spatial filter, and $\mathbf{P}_1$ and $\mathbf{P}_2$ are sample covariance matrices of two classes. The solution to (1) is obtained by the generalized EVD.

2.1. CSP with Spatio-Temporal Filtering

The traditional CSP algorithm uses only spatial filters to extract features. But since EEG signals are nonstationary, temporal filters can also be employed to improve classification accuracy. To this end, we shall develop an efficient algorithm to design spatial and temporal filters both based on the CSP technique.

Given data matrices $\{\mathbf{X}_i\}$, (1) is reformulated as follows to cope with spatio-temporal filtering

$$\max_{\mathbf{w} \in \mathbb{R}^C, \mathbf{h} \in \mathbb{R}^L} \frac{\mathbf{w}^T \mathbf{P}_1^h \mathbf{w}}{\mathbf{w}^T \mathbf{P}_2^h \mathbf{w}} \quad (2)$$

where $\mathbf{h} = [h_1 \ \dots \ h_L]^T$ represents the coefficient vector of a temporal filter, and $\{\mathbf{P}_k^h\}_{k=1}^2$ defined below are sample covariance matrices of temporally filtered EEG signals

$$\mathbf{P}_k^h = \frac{1}{|\mathcal{C}_k|} \sum_{i \in \mathcal{C}_k} \mathbf{X}_i \mathbf{H}^T \mathbf{H} \mathbf{X}_i^T, \text{ for } k = 1, 2. \quad (3)$$

In (3), $\mathcal{C}_k$ denotes the index set of data blocks associated with the k th mental state, and

$$\mathbf{H} = \begin{bmatrix} h_1 & \dots & h_L \\ & h_1 & \dots & h_L \\ & & \ddots & \ddots & \ddots \\ & & & h_1 & \dots & h_L \end{bmatrix}. \quad (4)$$

Eq. (4) indicates that different spatial channels share the same temporal filter. Thus, the number of filter coefficients designed for each class of EEG signals is $C + L$. Compared to the spatio-temporal filtering algorithm proposed in [11], where each channel is equipped with a temporal filter, yielding CL coefficients to be optimized, our model can reduce the overfitting risk especially in the case of small sample size.

2.2. Alternating Optimization

In practice, we adopt an alternating optimization algorithm to solve (2). More specifically, spatial and temporal filters are alternatively updated by fixing one and solving for the other. The iterative procedure continues until the updates of both filters are negligible or the maximal iteration T_1 is reached. More details are described below.

Given $\mathbf{h}^{(l-1)}$, (2) is tackled by the EVD to obtain $\mathbf{w}^{(l)}$. When updating the temporal filter, (2) is reformulated as

$$\max_{\mathbf{h} \in \mathbb{R}^L} \frac{\mathbf{h}^T \mathbf{P}_1^w \mathbf{h}}{\mathbf{h}^T \mathbf{P}_2^w \mathbf{h}}, \quad (5)$$

where

$$\mathbf{P}_k^w = \frac{1}{|\mathcal{C}_k|} \sum_{i \in \mathcal{C}_k} \mathbf{W}^{(l)} \hat{\mathbf{X}}_i \hat{\mathbf{X}}_i^T (\mathbf{W}^{(l)})^T, \quad k = 1, 2, \quad (6)$$

$$\mathbf{W}^{(l)} = \mathbf{I}_L \otimes (\mathbf{w}^{(l)})^T, \quad (7)$$

$$\hat{\mathbf{X}}_i = \begin{bmatrix} \mathbf{x}_{i,1} & \mathbf{x}_{i,2} & \dots & \mathbf{x}_{i,K-L+1} \\ \mathbf{x}_{i,2} & \mathbf{x}_{i,3} & \dots & \mathbf{x}_{i,K-L+2} \\ \vdots & \vdots & \dots & \vdots \\ \mathbf{x}_{i,L} & \mathbf{x}_{i,L+1} & \dots & \mathbf{x}_{i,K} \end{bmatrix}. \quad (8)$$

In (7), $\otimes$ denotes the Kronecker product.

2.3. Sparse Spatio-Temporal Filter Designs

The sparsity is an effective technique to improve the robustness and efficiency of the proposed CSP algorithm. Sparse spatial filters are able to identify EEG channels highly contaminated by noise and artifacts. Also, arithmetic operations corresponding to zero coefficients of temporal filters can be ignored, thus reducing the implementation complexity.

To achieve sparse filters, an ℓ_1 -norm regularization term is further incorporated in both (2) and (5). Given $\mathbf{h}$, (2) is modified to

$$\max_{\mathbf{w} \in \mathbb{R}^C} \frac{\alpha \mathbf{w}^T \mathbf{P}_1^h \mathbf{w} - (1 - \alpha) \|\mathbf{w}\|_1}{\mathbf{w}^T \mathbf{P}_2^h \mathbf{w}}, \quad (9)$$

Table 1: Classification accuracies of each algorithm

Subjects	CSP+LDA	Proposed+LDA	STRCSP+LDA	rCSP+LDA	ssCSP+LDA	MDRM
<i>aa</i>	83.91	94.66	84.39	84.06	84.06	70.06
<i>al</i>	98.71	99.11	97.54	93.89	93.89	95.27
<i>av</i>	72.30	77.89	73.45	73.76	73.76	54.38
<i>aw</i>	92.67	100	99.33	93.20	92.33	69.27
<i>ay</i>	97.67	93.33	90.50	98.00	97.67	61.17
mean±std	89.05 ± 11.04	92.99 ± 8.91	89.04 ± 10.55	88.58 ± 9.73	88.34 ± 9.55	70.03 ± 15.50
Subjects	CSP+SVM	Proposed+SVM	STRCSP+SVM	rCSP+SVM	ssCSP+SVM	MDRM
<i>aa</i>	83.68	89.86	87.41	84.52	84.46	-
<i>al</i>	98.97	99.15	97.67	95.22	95.22	-
<i>av</i>	68.11	74.79	74.16	73.50	73.73	-
<i>aw</i>	93.90	99.43	98.57	93.37	92.40	-
<i>ay</i>	97.67	96.50	85.00	97.17	97.17	-
mean±std	88.47 ± 12.86	91.95 ± 10.33	88.56 ± 10.06	88.76 ± 9.81	88.60 ± 9.62	-

where α is a regularization parameter used to control the relative importance of the sparsity of $\mathbf{w}$. To tackle the above problem, we reformulate $\|\mathbf{w}\|_1$ as $\|\mathbf{w}\|_1 = \mathbf{w}^T \Omega \mathbf{w}$ where Ω is a diagonal matrix whose i th diagonal entry is computed by $\Omega_{i,i} = \frac{1}{|w_i|}$. In practice, however, $\mathbf{w}$ are unknown variables and Ω cannot be computed beforehand. Thus, an iterative procedure nested in the previous alternating optimization algorithm is further introduced to update spatial filter coefficients. At each inner iteration, (9) is cast as

$$\mathbf{w}^{(l,s)} = \arg \max_{\mathbf{w} \in \mathbb{R}^C} \frac{\mathbf{w}^T [\alpha \mathbf{P}_1^h - (1 - \alpha) \Omega^{(l,s-1)}] \mathbf{w}}{\mathbf{w}^T \mathbf{P}_2^h \mathbf{w}}, \quad (10)$$

where the diagonal element of $\Omega^{(l,s-1)}$ is computed by

$$\Omega_{i,i}^{(l,s-1)} = \left(\eta + |w_i^{(l,s-1)}| \right)^{-p}. \quad (11)$$

Here, $w_i^{(l,s-1)}$ is the i th spatial filter coefficient obtained in the previous inner iteration. Positive constant η (e.g., 10^{-8} used in our experiments) is introduced to avoid division by zero, and power $p \geq 1$ is used to accelerate the convergence. Problem (10) can still be viewed as a CSP problem and efficiently solved by the generalized EVD. Its solution is taken into (11) to update $\Omega^{(l,s)}$. The inner iterative procedure is terminated when the update of $\mathbf{w}^{(l,s)}$ is negligible or the specified maximal iteration T_2 (e.g., $T_2 = 500$) is reached. Once the inner iterative procedure converges, the final solution is returned as $\mathbf{w}^{(l)}$ to the outer iterative procedure.

The same iterative reweighting algorithm described above is also employed to sparsify temporal filters. Specifically, a regularized (5) is solved in each inner iteration

$$\mathbf{h}^{(l,s)} = \arg \max_{\mathbf{h} \in \mathbb{R}^L} \frac{\mathbf{h}^T [\alpha \mathbf{P}_1^w - (1 - \alpha) \Phi^{(l,s-1)}] \mathbf{h}}{\mathbf{h}^T \mathbf{P}_2^w \mathbf{h}}. \quad (12)$$

Diagonal matrix $\Phi^{(l,s-1)}$ is defined similar to $\Omega^{(l,s-1)}$ by replacing $w_i^{(l,s-1)}$ in (11) by $h_i^{(l,s-1)}$.

3. EXPERIMENTAL RESULTS

3.1. Experimental Setup

The data used in our experiments are taken from Dataset IVa of BCI Competition III [20]. EEG signals are recorded through 118 channels with sampling frequency 100 Hz, which includes 2 MI tasks (i.e., right hand, foot associated with labels 1 and 2, respectively). EEG signals in each trial are extracted from the period of 0.5 ~ 3.5s after a visual cue, and then filtered by a bandpass filter with passband 9 ~ 35 Hz.

3.2. Performance Comparison of Existing Algorithms

Classification accuracies of the proposed algorithm are compared with those of five state-of-the-arts methods, including traditional CSP, spatio-temporally regularized CSP (STRCSP) [11], rCSP [13]-[14], ssCSP [15] and MDRM [18]. Except the MDRM, the other feature extraction methods work along with both LDA and SVM in our evaluation. Furthermore, 10×10 -fold cross validation is employed in our experiment.

Table 1 lists classification accuracies for five subjects *aa*, *al*, *av*, *aw* and *ay*. The means and standard deviations of classification accuracies of five subjects are also given for each algorithm. The best results are highlighted in bold font for each subject. It can be observed that the proposed approach equipped with either LDA or SVM achieves the best results in the experiments of four subjects, while the rCSP with the LDA classifier outperforms the other approaches only in the experiment of subject *ay*. In average, our algorithm using the LDA classifier yields, respectively, 3.94%, 3.95%, 4.41%, 4.65% and 22.96% performance improvement over the other five approaches. Similar results are also achieved by the proposed method with the SVM classifier.

Fig. 1 shows magnitude responses of a pair of temporal filters designed for subjects *aa*, *av* and *aw*. Although features in MI tasks are generally subject-independent, temporal

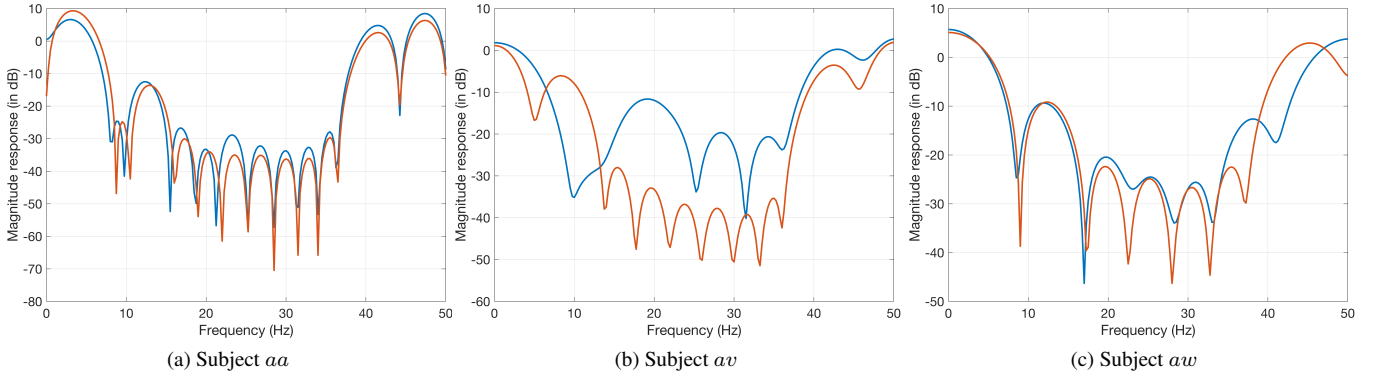


Fig. 1: Magnitude responses of temporal filters designed for subjects *aa*, *av* and *aw*.

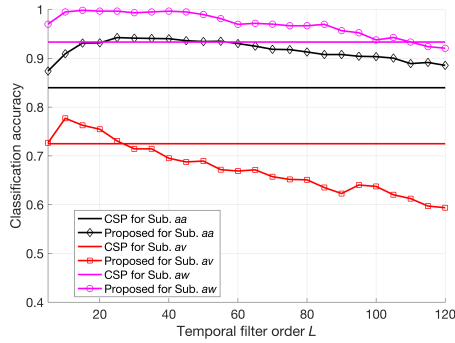


Fig. 2: Variation of classification accuracies w. r. t. L .

filters for different subjects show similar spectral characteristics, that suppress spectral components within $20 \sim 30$ Hz.

3.3. Effects of L and α

In the proposed algorithm, temporal filter order L and regularization parameter α need to be chosen appropriately. Two sets of experiments are conducted in this subsection to evaluate their impacts on classification accuracy. Only LDA is used as the classifier.

First of all, temporal filter order L varies from 5 to 120, when $\alpha = 0.1$. In this experiment, we only present experimental results of subjects *aa*, *av* and *aw*. Fig. 2 shows the variation of classification accuracies with respect to L . Experimental results of traditional CSP are also given as baselines. It can be observed that, with the increase of L , classification performance is often improved first and then gradually becomes worse. But, for different subjects, the best result is generally achieved by different L for different subjects.

Then, we fix $L = 15$ and vary α from 0.05 to 1 with incremental step 0.05. Variations of classification accuracies and also sparsities of spatial and temporal filters of subject *aw* with respect to α are illustrated in Fig. 3. Here, the sparsity of spatial or temporal filters counts the total number of zero

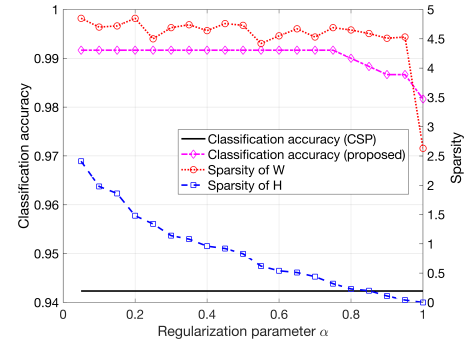


Fig. 3: Variation of classification accuracies and sparsities of filter coefficients w. r. t. α for subject *aw*.

coefficients in the same class of filters. It can be noticed that the fluctuation of classification accuracies is very flat, even though more coefficients are nullified with the decrease of α . This implies that α has a limited impact on final results.

4. CONCLUSIONS

In this paper, an efficient CSP algorithm using spatio-temporal filtering has been developed for EEG-based MI classification. Compared to traditional joint filtering strategies, our approach designs a common temporal filter for EEG signals of all the spatial channels to enhance classification performance and also avoid the overfitting risk often encountered in the case of small sample size. An alternating optimization scheme has been developed for the optimization of both spatial and temporal filters. Furthermore, the ℓ_1 -norm regularization is introduced in both spatial and temporal filters to improve sparsity. Using the reweighting technique, the resulting sparse filter design problem in each iteration can still be formulated in the classical CSP form and efficiently solved by the generalized EVD. Experimental results have demonstrated the superior performance of the proposed algorithm over the state-of-the-arts methods.

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